AEO "Nazarbayev Intellectual Schools"
Cambridge International Examinations

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CENTRE NUMBER						CANDIDATE NUMBER				

0759115582

CHEMISTRY **Grade 12**

Paper 2 **May 2018**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Calculator
Data Booklet

12CHEM/02

READ THESE INSTRUCTIONS FIRST

Write your name, centre number and candidate number in the spaces at the top of the page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
You may receive fewer marks if you do not show your working or do not use appropriate units where required.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 100.

Answer all questions in English.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
Total	

This document consists of **18** printed pages and **2** blank pages.

CAMBRIDGE International Examinations

1 Several complex ions contain the Co^{2+} ion. Two such complex ions are $[Co(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$.

For Examiner's Use

(a) In the $[Co(H_2O)_6]^{2+}$ complex ion the Co^{2+} ion is surrounded by six water molecules, which act as ligands.

Explain what is meant by a *ligand*.

.... [1]

(b) **(i)** Draw the three dimensional shape of the $[Co(H_2O)_6]^{2+}$ complex ion.

2+

Co

[1]

(ii) Name the shape of this complex ion.

.... [1]

(c) **(i)** Draw the three dimensional shape of the $[CoCl_4]^{2-}$ complex ion.

2–

Co

[1]

(ii) Name the shape of this complex ion.

.... [1]

For Examiner's Use

(d) The $[CoCl_4]^{2-}$ complex ion is formed when excess concentrated hydrochloric acid is added to a solution of the $[Co(H_2O)_6]^{2+}$ complex ion. In this reaction the colour of the solution changes from pink to blue.

(i) Write a balanced chemical equation for this reaction. State symbols are **not** necessary.

.... [1]

(ii) Explain why a solution of the $[Co(H_2O)_6]^{2+}$ complex ion is coloured.

....

....

....

....

.... [3]

(iii) Explain why a solution of the $[Co(H_2O)_6]^{2+}$ complex ion is pink but a solution of the $[CoCl_4]^{2-}$ complex ion is blue.

....

....

....

....

.... [2]

[Total: 11]

For Examiner's Use

2 A chemist constructs an electrochemical cell using a VO_2^+ / VO^{2+} half-cell and a Zn^{2+} / Zn half-cell.

(a) Draw a labelled diagram of the electrochemical cell that he constructs. Your diagram should include:

- two beakers
- a method for measuring the voltage of the cell
- the names of all substances used in solution
- the names of all substances used as solids
- the names of any other equipment used
- details of all concentrations and conditions used.

[5]

(b) Calculate the voltage of the electrochemical cell under standard conditions. Include in your answer any relevant data you have used.

.... V [2]

(c) Deduce the balanced chemical equation for the reaction that takes place when the electrochemical cell is in use.

For Examiner's Use

.... [2]

(d) Deduce the oxidation state of vanadium in VO_2^+ and in VO^{2+}.

oxidation state of vanadium in VO_2^+ ....

oxidation state of vanadium in VO^{2+} .... [1]

[Total: 10]

For Examiner's Use

3 Element 16, sulfur, has four stable isotopes. These isotopes and their relative abundances are shown.

isotope	**relative abundance**
^{32}S	95.02%
^{33}S	0.75%
^{34}S	4.21%
^{36}S	0.02%

Sulfur forms ionic and covalent compounds.

Some of its ions are:

sulfide ion, S^{2-} sulfate ion, SO_4^{2-} thiosulfate ion, $S_2O_3^{2-}$

(a) Deduce the number of protons, neutrons and electrons in one $^{36}S^{2-}$ ion.

protons ....

neutrons ....

electrons ....

[1]

(b) Use the data in the table to calculate the relative atomic mass of sulfur.
Show **all** your working.
Give your answer correct to **four** significant figures.

.... [2]

(c) In each thiosulfate ion, $S_2O_3^{2-}$, one of the sulfur atoms is central and is bonded covalently to the four other atoms. This central sulfur atom has twelve electrons on its outer shell. The four other atoms are not directly bonded to each other. Each of these four atoms has eight electrons on its outer shell.

Suggest a dot-and-cross diagram for the thiosulfate ion, $S_2O_3^{2-}$.

[2]

For Examiner's Use

(d) Sodium sulfide, Na_2S, has a giant ionic structure. Hydrogen sulfide, H_2S, has a simple molecular (simple covalent) structure.

(i) Describe and explain the different electrical conductivities of sodium sulfide **and** hydrogen sulfide.

....

....

....

....

.... [3]

(ii) Describe and explain the difference in the melting points of sodium sulfide **and** hydrogen sulfide.

....

....

....

....

.... [3]

(e) **(i)** Explain why sulfur is **less** electronegative than chlorine (element 17).

....

.... [1]

(ii) Explain why sulfur is **more** electronegative than selenium (element 34). Answer as fully as you can.

....

....

....

....

.... [3]

[Total: 15]

For Examiner's Use

4 Nitrogen will react with hydrogen to form ammonia.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H^{\ominus} = -92\,\text{kJ mol}^{-1}$$

(a) Use this information and data from the Data Booklet to calculate an average value for the N–H bond energy in ammonia.

[3]

(b) Predict whether the entropy change for the forward reaction

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

has a positive sign or a negative sign. Explain your answer.

....

....

.... [1]

(c) The forward reaction $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ is spontaneous at temperatures below 463 K. It is not spontaneous at temperatures above 463 K.

Calculate the entropy change, ΔS, for the forward reaction

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

Include units in your answer.

.... units .... [4]

For Examiner's Use

(d) A chemist dissolved 400 cm^3 of ammonia gas, measured **under room conditions**, in water at 18.0 °C. When the ammonia was completely dissolved, the temperature of the solution was 24.0 °C. The final mass of solution obtained was 20.0 g.

(i) Calculate the number of joules of heat energy released during the experiment. You should assume that the specific heat capacity of the solution produced is $4.18\ J\ g^{-1}\ K^{-1}$.

.... J [2]

(ii) Calculate the number of moles of ammonia used in the experiment.

.... mol [1]

(iii) Calculate the enthalpy change of solution of ammonia.

.... $kJ\ mol^{-1}$ [1]

[Total: 12]

5 (a) (i) Give an expression for the ionic product of water.

.... [1]

(ii) The numerical value of the ionic product of water is $1.00 \times 10^{-14}\,mol^2\,dm^{-6}$. Use this value and your answer to **(a)(i)** to show why the pH of pure water is 7.00 at 298 K.

[2]

(b) The pH of pure water at its boiling point, 373 K, is pH = 6.14.

(i) Calculate the hydrogen ion concentration, $[H^+]$, in pure water at 373 K.

.... [1]

(ii) Give the hydroxide ion concentration, $[OH^-]$, in pure water at 373 K.

.... [1]

(iii) The $[H^+]$ is different at 298 K and 373 K. Suggest a reason for this. Explain your answer.

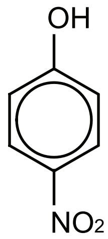
....

....

.... [2]

For Examiner's Use

(c) The pK_a of 4-nitrophenol is 7.15.



4-nitrophenol

A buffer solution is made by adding 0.200 moles of 4-nitrophenol to a solution containing 0.100 moles of sodium hydroxide dissolved in 2000 cm^3 of water. The 0.100 moles of sodium hydroxide react with 0.100 moles of the 4-nitrophenol.

$$NO_2C_6H_4OH + NaOH \rightarrow NO_2C_6H_4O^{-}Na^{+} + H_2O$$

(i) Identify the substance that acts as an acid and the substance that acts as a base in this reaction. Explain your answer.

acid ....

reason ....

base ....

reason .... [1]

(ii) Calculate the pH of the buffer solution.

pH = ... [2]

(iii) The buffer solution is mixed with 0.005 moles of pure HNO_3.
Calculate the pH of the buffer solution after the addition of the HNO_3.
Show your working.

pH = ... [3]

[Turn over

For Examiner's Use

(iv) Explain how your answers to **(c)(ii)** and **(c)(iii)** show that this is a **buffer** solution.

.... [1]

[Total: 14]

6 Butane, C_4H_{10} , and but-2-ene, C_4H_8 , both react with chlorine gas. The two reactions have different conditions, different mechanisms and give different products.

(a) Butane reacts with chlorine if the two substances are mixed together in the presence of ultra-violet light. The mechanism is free-radical substitution. It is known as photochemical chlorination.

(i) State what is meant by a *free radical*.

.... [1]

(ii) Explain, with the aid of an equation, the role of ultra-violet light in this reaction.

....

.... [2]

(iii) The first, initiation, step in this reaction mechanism involves bond breaking.

Name the type of bond breaking involved.

.... [1]

(iv) The reaction mechanism involves a number of different possible termination steps.

Write a balanced chemical equation for one possible termination step. You should use structural formulae for any organic species involved.

[1]

For Examiner's Use

(b) But-2-ene reacts with bromine monochloride, BrC*l*, if the two substances are mixed together in the dark. The mechanism is electrophilic addition.

(i) Write a balanced chemical equation for this reaction using structural formulae for the two organic substances.

[1]

(ii) Identify the substance that behaves as an electrophile in this reaction and explain why it can be classified as an electrophile.

....

.... [2]

(iii) Name the organic product of this reaction.

.... [1]

(iv) Show, by reference to its displayed formula, why this product is chiral. You should clearly identify all chiral carbon atoms.

[2]

(c) But-2-ene exists as a pair of geometrical isomers. Draw and name these two isomers. Each name should be written underneath the compound it refers to.

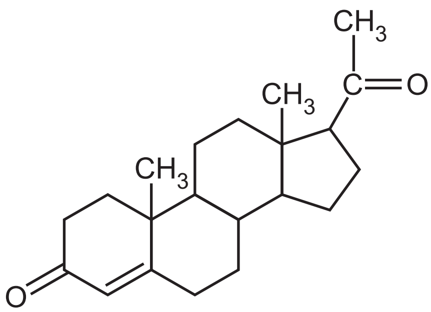
isomer 1	isomer 2
name	name ....

[2]

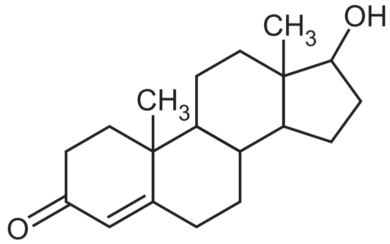
[Total: 13]

[Turn over

7 Progesterone is a female sex hormone. Testosterone is a male sex hormone.



progesterone



testosterone

(a) Describe one chemical test that would distinguish a sample of progesterone from a sample of testosterone.

testing reagent ....

visible result with progesterone ....

visible result with testosterone ....

[3]

(b) Describe the results seen when warm Tollens' reagent is added separately to each hormone. If the addition of Tollens' reagent causes no change you should write 'no change'.

progesterone + warm Tollens' reagent ....

testosterone + warm Tollens' reagent ....

[2]

(c) **(i)** Draw the skeletal formulae of the products formed when an ethanolic solution of sodium tetraborohydride, $NaBH_4$, is added separately to each hormone.

product of progesterone + $NaBH_4$

product of testosterone + $NaBH_4$

[2]

For Examiner's Use

For Examiner's Use

(ii) The mechanism of the reaction between testosterone and $NaBH_4$ is the same as the mechanism of the reaction between propanone and HCN.

Suggest the identity of the first species that attacks testosterone when testosterone reacts with $NaBH_4$.

.... [1]

[Total: 8]

8 Propanoic acid is a carboxylic acid.

(a) A solution is made by dissolving 1.48 g of propanoic acid in sufficient water. An excess of sodium carbonate is then added.

(i) Write a balanced chemical equation for the reaction that occurs.

.... [1]

(ii) Name the organic product of the reaction.

.... [1]

(iii) Calculate the volume of gas produced **under room conditions**.

.... dm^3 [1]

(b) Propanoic acid can be made using a range of different chemical reactions.

For each of the reactions that follow you should name all reactants, organic and inorganic, and describe any essential reaction conditions.

(i) Describe how propanoic acid can be made using an oxidation reaction.

reactants ....

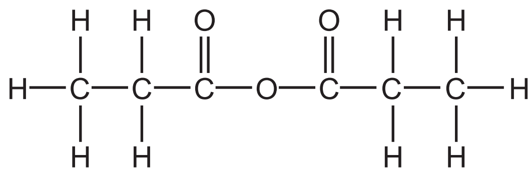
conditions .... [2]

(ii) Describe how propanoic acid can be made using a hydrolysis reaction.

reactants ....

conditions .... [2]

(c) Propanoic acid can be used to make propanoic anhydride. Propanoic anhydride is important in organic synthesis.



propanoic anhydride

Name the two compounds formed when propanoic anhydride is reacted with an **excess** of ammonia.

....

.... [2]

[Total: 9]

For Examiner's Use

For Examiner's Use

9 A chemist wishes to analyse a protein and discover the sequence of amino acids it consists of.
She partially hydrolyses the protein. Amongst the products are two peptides, which she calls **A** and **B**.

A and **B** both contain eight amino acid residues.

(a) She subjects peptide **A** to further hydrolysis. The hydrolysis products include only five different tripeptides. These tripeptides are:

gly – ser – ser
lys – ala – lys
ser – lys – ala
ser – ser – lys
ser – ser – ser

Deduce the amino acid sequence of peptide **A**.

.... [3]

(b) The sequence of peptide **B** is:

asp – val – ser – thr – phe – ile – glu – asp

In aqueous solution it forms a globular shape. Three of the amino acid residues have side chains (R-groups), which are positioned on the inside of the globule.

Identify the three amino acid residues with side chains positioned on the inside of the globule. Explain your answer.

....

....

.... [3]

For Examiner's Use

(c) The R-group of the amino acid lysine is $(CH_2)_4NH_2$.

(i) Draw the displayed formula of lysine at pH 14.

[1]

(ii) Draw the displayed formula of lysine at pH 1.

[1]

[Total: 8]

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